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Revision Number	Revision Date	Revision Reason
1.0	24.12.2025	First Issue

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1. PURPOSE

This project aims to develop a machine learning model to automatically predict passenger comfort level based on cabin environmental parameters and IFE system usage data. Inputs include variables such as temperature, humidity, seat location, noise level, air flow, screen brightness, and IFE usage duration. The model outputs a comfort index (1–5 scale). AI Technique: Supervised Learning – Regression / Classification (Random Forest)

2. SCOPE

Aerospace Domain: Cabin Systems – In-Flight Entertainment and Environmental Control. The model can be used to support optimization of cabin and IFE system parameters to enhance passenger comfort.

3. SYSTEM MODELING AND THEORETICAL BACKGROUND

3.1. AI Model Architecture Model Type

Random Forest Regressor (baseline) or Neural Network. Inputs: Temperature, Humidity, Cabin Noise, Airflow, Seat Location, IFE Screen Brightness, IFE Usage Time. Output: Comfort Index (1–5 scale). Learning Objective: Minimize Mean Squared Error (MSE) and maximize R^2 score.

3.2. Physical System Integration System: Aircraft Cabin & IFE Environment AI Interaction

The model uses cabin sensor data or system parameters (temperature, humidity, brightness, etc.) to predict passenger comfort. The predictions can provide feedback to adjust environmental control or IFE settings automatically.

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3.3. Data and Preprocessing

The 1000 row dataset is simulated using Python (NumPy, Pandas) Synthetic variables like IFE brightness and screen usage time can be added for system relevance.

Variable	Description	Unit
Temperature	Cabin air temperature	°C
Humidity	Relative humidity	%
Noise	Sound level	dB
Airflow	Ventilation rate	m ³ /h
SeatZone	Cabin zone (Front/Mid/Aft)	Categorical
IFE_Brightness	Screen brightness level	%
IFE_UsageTime	Screen usage duration	min
ComfortIndex	Passenger comfort level	1-5

Figure 1 Key Variables

Data Processing:

- Missing values filled with mean
- Normalization (Min–Max)
- One-hot encoding for categorical variables
- Correlation analysis and feature importance visualization

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3.4. Software Implementation

Python Libraries:

- pandas, numpy for data preprocessing
- scikit-learn for Random Forest.

3.5. Visualization

- Correlation heatmap
- Predicted vs Actual comfort scatter plot
- Feature importance plot

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4. WORK STAGES

4.1. Library Imports

This section incorporates the essential Python libraries required for data processing, mathematical computations, visualization, and machine learning into the project.

Table 1 Library Definition

Code Module	Library / Tool	Technical Description
<code>import pandas as pd</code>	Pandas	Used for managing and analyzing structured data through DataFrames.
<code>import numpy as np</code>	NumPy	Supports large, multi-dimensional arrays and matrices; heavily used for generating synthetic data.
<code>import matplotlib.pyplot as plt</code>	Matplotlib	The fundamental library for creating static, interactive, and animated visualizations.
<code>import seaborn as sns</code>	Seaborn	A high-level library built on Matplotlib for drawing attractive and informative statistical graphics.
<code>from sklearn.model_selection import ...</code>	Model Selection	Provides <code>train_test_split</code> for data partitioning and <code>GridSearchCV</code> for exhaustive hyperparameter optimization.
<code>from sklearn.ensemble import ...</code>	Ensemble Learning	Imports the Random Forest Regressor , a popular ensemble algorithm used for regression tasks.
<code>from sklearn.preprocessing import ...</code>	Preprocessing	Used for data normalization (<code>MinMaxScaler</code>) and converting

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Code Module	Library / Tool	Technical Description
		categorical variables into numerical format (<code>OneHotEncoder</code>).
<pre>from sklearn.compose import ...</pre>	Column Transformer	A utility to apply different preprocessing steps (scaling vs. encoding) to specific columns in the dataset.
<pre>from sklearn.metrics import ...</pre>	Performance Metrics	Imports evaluation tools such as Mean Squared Error (MSE) and R-squared (R^2) to assess model accuracy.

4.2. Synthetic Data Generation (NumPy & Pandas)

This stage involves creating a controlled, simulated dataset to model passenger comfort. By using NumPy for mathematical randomness and Pandas for structured storage, we generate realistic scenarios that mimic actual cabin environments.

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4.2.1. Defining of Input Variables

Table 2 Defining of Input Variables

Code Snippet	Variable / Action	Technical Description
<code>np.random.seed(42)</code>	Random Seed Initialization	Sets a specific seed for the NumPy random number generator. This ensures reproducibility , allowing the same dataset to be generated every time the code is executed.
<code>N_samples = 1000</code>	Sample Size Definition	Defines the total number of observations (rows) to be created in the dataset (1000 samples).
<code>Temperature = np.random.uniform(18, 26, ...)</code>	Uniform Distribution (Temp)	Generates 1000 random temperature values between 18°C and 26°C using a uniform distribution , ensuring a balanced spread across the range.
<code>SeatZone_map = {0: 'Front', ...}</code>	Categorical Mapping	Creates a dictionary to map numerical integer values to their corresponding textual categories ('Front', 'Mid', 'Aft') for the SeatZone feature.
<code>SeatZone = np.random.randint(0, 3, ...)</code>	Integer Sampling	Randomly selects 1000 integers from the range [0, 1, 2]. These serve as the underlying indices for the SeatZone categories.
Other Input Variables	Feature Synthesis	Similarly, 1000 random samples are generated for Humidity, Noise, Airflow, IFE Brightness, and IFE Usage Time using specific ranges to mimic a real cabin environment.

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4.2.2. Target Variable (Comfort Index) Calculation

Table 3 Target Variable Calculation

Code Snippet	Variable / Action	Technical Description
<code>ComfortIndex_raw = (...)</code>	Baseline Regression Formula	This is the core mathematical formula defining how input variables affect the Comfort Index . For example, it models a decrease in comfort as temperature deviates from the 22°C ideal.
<code>np.random.normal(0, 0.5, ...)</code>	Stochastic Noise Injection	Adds 1,000 random values from a normal distribution (mean=0, std=0.5). This simulates the "noise" or unpredictable factors found in real-world data, preventing the model from learning an overly perfect pattern.
<code>np.clip(ComfortIndex_raw, 1, 5)</code>	Value Constraint (Clipping)	Constrains the raw comfort scores to a specific range between 1 and 5 . This ensures no data points fall outside the logical boundaries of the comfort scale.
<code>.round(0)</code>	Discretization	Rounds the values to the nearest integer. This transforms the continuous output into an ordinal scale (1 to 5 stars), mimicking a typical passenger rating system.

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4.2.3. DataFrame Construction

Table 4 DataFrame Construction

Code Snippet	Variable / Action	Technical Description
<code>pd.DataFrame({...})</code>	DataFrame Creation	Orchestrates the individual arrays (Temperature, Humidity, etc.) into a cohesive tabular structure where each column represents a feature and each row represents a passenger observation.
Feature Columns	Temperature, Humidity, Noise, etc.	These serve as the Independent Variables (X) that the model will use to learn patterns.
Target Column	<code>ComfortIndex</code>	This is the Dependent Variable (y) that the model aims to predict.
<code>df.head()</code>	Data Preview	Displays the first few rows of the constructed dataset to verify that the data generation and mapping (e.g., SeatZone categories) were successful.

4.3. Data Splitting and Preprocessing

This stage involves preparing the data for the machine learning model.

4.3.1. Feature and Target Separation

Before training, the data is separated into independent variables (features) and the dependent variable (target), then split into subsets for learning and evaluation.

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Table 5 Feature and Target Separation

Code Snippet	Technical Description
<pre>X = data.drop('ComfortIndex', axis=1)</pre>	Creates the Feature Set (X) by removing the target column from the DataFrame (<code>axis=1</code> denotes column-wise operation).
<pre>y = data['ComfortIndex']</pre>	Defines the Target Variable (y) , which represents the values the model aims to predict.
<pre>X_train, X_test, y_train, y_test = train_test_split(...)</pre>	Splits the dataset into 80% Training (for model learning) and 20% Testing (for independent performance evaluation).
<pre>test_size=0.2, random_state=42</pre>	Reserves 20% of data for testing and fixes the random seed to ensure reproducibility across different runs.

4.3.2. Building the Preprocessing Pipeline

Using a ColumnTransformer is a best practice that allows different mathematical transformations to be applied to specific column types (numerical vs. categorical).

Table 6 Building the Preprocessing Pipeline

Code Snippet	Technical Description
<pre>categorical_features = ['SeatZone']</pre>	Identifies columns containing text-based categories to be processed via encoding.
<pre>numerical_features = [...]</pre>	Lists all continuous numerical columns that require normalization/scaling.
<pre>ColumnTransformer(...)</pre>	Constructs a transformation engine that directs specific features to their respective processors.

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Code Snippet	Technical Description
<pre>('num', MinMaxScaler(), numerical_features)</pre>	Numerical Scaling: Applies <code>MinMaxScaler</code> to normalize all numerical values to a range between 0 and 1 .
<pre>('cat', OneHotEncoder(...), categorical_features)</pre>	Categorical Encoding: Converts text labels (e.g., 'Front') into Binary (0 or 1) columns so the model can process them.
<pre>remainder='passthrough'</pre>	Ensures any columns not explicitly mentioned are passed through the pipeline without modification.

4.3.3. Executing the Preprocessing

This phase applies the designed pipeline to the data. It is critical to follow the "Fit-Transform" logic to prevent data leakage.

Table 7 Executing the Preprocessing

Code Snippet	Technical Description
<pre>X_train_processed = preprocessor.fit_transform(X_train)</pre>	The pipeline learns parameters (like min/max values) from the training set and then transforms it.
<pre>X_test_processed = preprocessor.transform(X_test)</pre>	The pipeline only transforms the test set using parameters learned from the training set. It does <i>not</i> re-learn from the test data.
<pre>feature_names = ...</pre>	Extracts the new column names generated during transformation (especially those created by the One-Hot Encoder).

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Code Snippet	Technical Description
<pre>X_train_df = pd.DataFrame(...)</pre>	Converts the processed NumPy arrays back into a labeled Pandas DataFrame for better readability and tracking.

4.4. Data Discovery (Correlation Analysis)

This section creates a **Correlation Heatmap** to visualize the linear relationships between numerical variables.

Table 8 Correlation Analysis

Code Snippet	Technical Description
<pre>numerical_data = data.select_dtypes(include=np.number)</pre>	Filters the dataset to include only numerical columns, as correlation calculations require numeric inputs.
<pre>plt.figure(figsize=(10, 8))</pre>	Sets the dimensions of the visualization for better readability.
<pre>sns.heatmap(numerical_data.corr(), annot=True, ...)</pre>	Draws the Heatmap: Calculates the correlation matrix, displays the values on the chart (<code>annot=True</code>), and applies the <code>coolwarm</code> color palette (positive: warm, negative: cool).
<pre>plt.title(...) / plt.show()</pre>	Assigns a title to the visualization and renders the plot.

4.5. Model Training and Optimization

In this section, the **Random Forest Regressor** is defined, hyperparameters are optimized, and the final model is trained.

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4.5.1. Model Selection

Table 9 Model Selection

Code Snippet	Technical Description
<pre>rfr = RandomForestRegressor(random_state=42)</pre>	Initializes the Random Forest Regressor. This is a powerful ensemble learning algorithm that makes predictions by combining the outputs of multiple decision trees.

4.5.2. Hyperparameter Optimization

Table 10 Hyperparameter Optimization

Code Snippet	Technical Description
<pre>param_grid = { ... }</pre>	A dictionary of hyperparameter values for GridSearchCV to test: n_estimators (number of trees), max_depth (tree depth), and min_samples_leaf (minimum samples per leaf).
<pre>grid_search = GridSearchCV(...)</pre>	Creates the Grid Search Cross-Validation object to test all parameter combinations using 3-fold cross-validation (cv=3) and Mean Squared Error (MSE) as the scoring metric.
<pre>grid_search.fit(X_train_df, y_train)</pre>	Executes the grid search on the training data to identify the optimal parameter combination.
<pre>best_rfr = grid_search.best_estimator_</pre>	Assigns the best-performing model instance found by the search to a variable.
<pre>print(...)</pre>	Displays the best hyperparameters and the corresponding best MSE score achieved.

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4.6. Model Evaluation

The performance of the trained model is measured and visualized using the independent test dataset.

Table 11 Model Evaluation

Code Snippet	Technical Description
<pre>y_pred = best_rfr.predict(X_test_df)</pre>	Uses the optimized model to generate target predictions (y_pred) for the unseen test data.
<pre>mse = mean_squared_error(y_test, y_pred)</pre>	Calculates the Mean Squared Error (MSE) . This metric penalizes larger errors and should be minimized.
<pre>r2 = r2_score(y_test, y_pred)</pre>	Calculates the R-squared (R²) score, representing the proportion of variance explained by the model (ranges from 0 to 1; higher is better).
<pre>print(...)</pre>	Prints the calculated performance metrics in a formatted output.

4.6.1. Actual vs. Predicted Values Plot

Table 12 Actual vs. Predicted Values Plot

Code Snippet	Technical Description
<pre>plt.scatter(y_test, y_pred, alpha=0.6)</pre>	Creates a scatter plot with actual values on the X-axis and predicted values on the Y-axis.
<pre>plt.plot([y.min(), y.max()], ..., 'r--')</pre>	Draws a Perfect Prediction Line (y=x) . The proximity of the dots to this red dashed line indicates the model's accuracy.

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4.6.2. Feature Importance Ranking Plot

Table 13 Feature Importance Ranking Plot

Code Snippet	Technical Description
<pre>feature_importances = pd.Series(...)</pre>	Extracts the importance scores calculated by the Random Forest model, showing how much each input feature contributed to the final prediction.
<pre>sns.barplot(...)</pre>	Renders a bar chart sorting features from most to least important. This helps identify which variables (e.g., Temperature, Noise) the model relies on most.

5. RESULTS

5.1. Correlation Heat Map

The heatmap illustrates the strength and direction (positive or negative) of the relationships between all numerical variables in the dataset. The correlation coefficient ranges from **-1.00** to **+1.00**. The primary focus should be on the **ComfortIndex** row/column.

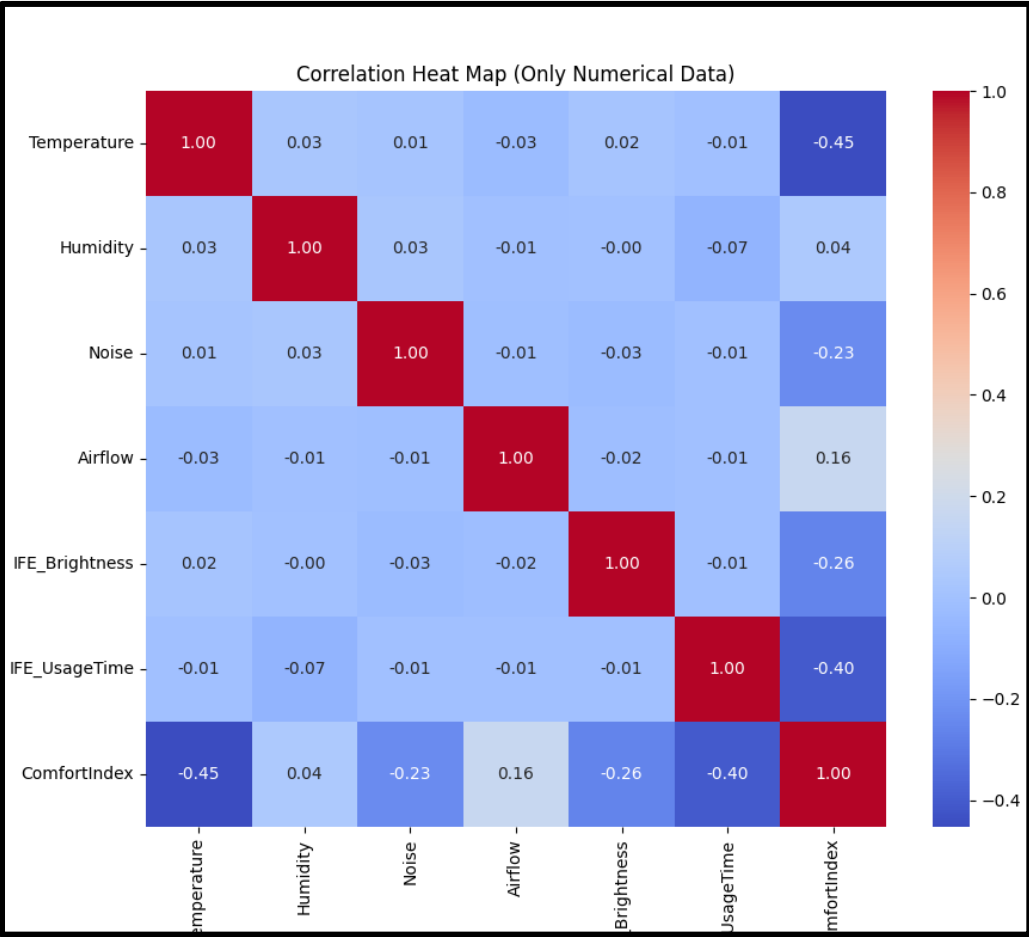


Figure 2 Correlation Heat Map

Table 14 Correlation Analysis Description

Correlation Value	Visual Representation	Interpretation
+1.00 (Deep Red)	Perfect Positive Correlation	As one variable increases, the other increases at the same rate. (Values on the diagonal are always 1.00 as they represent a variable's correlation with itself).

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Correlation Value	Visual Representation	Interpretation
-1.00 (Deep Blue)	Perfect Negative Correlation	As one variable increases, the other decreases at the same rate.
0.00 (White/Light Blue)	No Correlation	No linear relationship exists between the variables.

Table 15 Correlation Analysis

Feature	Correlation Value	Color	Comment
Temperature	-0.45	Red/Burgundy (Strong Negative)	Moderate-Strong Negative Correlation. As temperature increases, the Comfort Index decreases. This aligns with the logic that comfort drops as temperature deviates from 22°C.
IFE_UsageTime	-0.40	Blue (Moderate Negative)	Moderate Negative Correlation. As In-Flight Entertainment (IFE) usage time increases, the Comfort Index decreases, assuming long usage leads to discomfort.
IFE_Brightness	-0.26	Blue (Weak Negative)	Weak Negative Correlation. As IFE brightness increases, the Comfort Index slightly decreases.
Noise	-0.23	Blue (Weak Negative)	Weak Negative Correlation. As noise levels increase, the Comfort Index slightly decreases.
Airflow	0.16	Light Blue/White (Very Weak Positive)	Very Weak Positive Correlation. A slight increase in airflow leads to a marginal increase in the Comfort Index.

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Humidity	0.04	White (Negligible)	Almost no correlation. Since the formula uses an absolute value $\text{abs}()$ for deviation from 45% humidity, a simple linear correlation cannot capture the relationship. This non-linear pattern will be harder for a linear model to learn.
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- **Most Significant Linear Effects:** *Temperature* and *IFE_UsageTime* are the two variables that have the strongest linear impact on the **Comfort Index**.

Consequently, we can expect the model to assign high feature importance to these variables.

- **Multicollinearity (Inter-Feature Correlation):** Examining the relationships between input features (visible in the upper-left section of the correlation matrix), all values are near **0.00** (e.g., the correlation between *Temperature* and *Humidity* is 0.03). This indicates that your input variables are largely **independent** of one another. This is an ideal scenario for training a regression model, as it allows the model to clearly distinguish the individual contribution of each feature to the final prediction.

5.2. Predicted vs. Actual Comfort Plot

This visualization is used to evaluate the predictive accuracy of the **Random Forest Regressor** model by comparing its estimates against the ground truth labels in the test set.

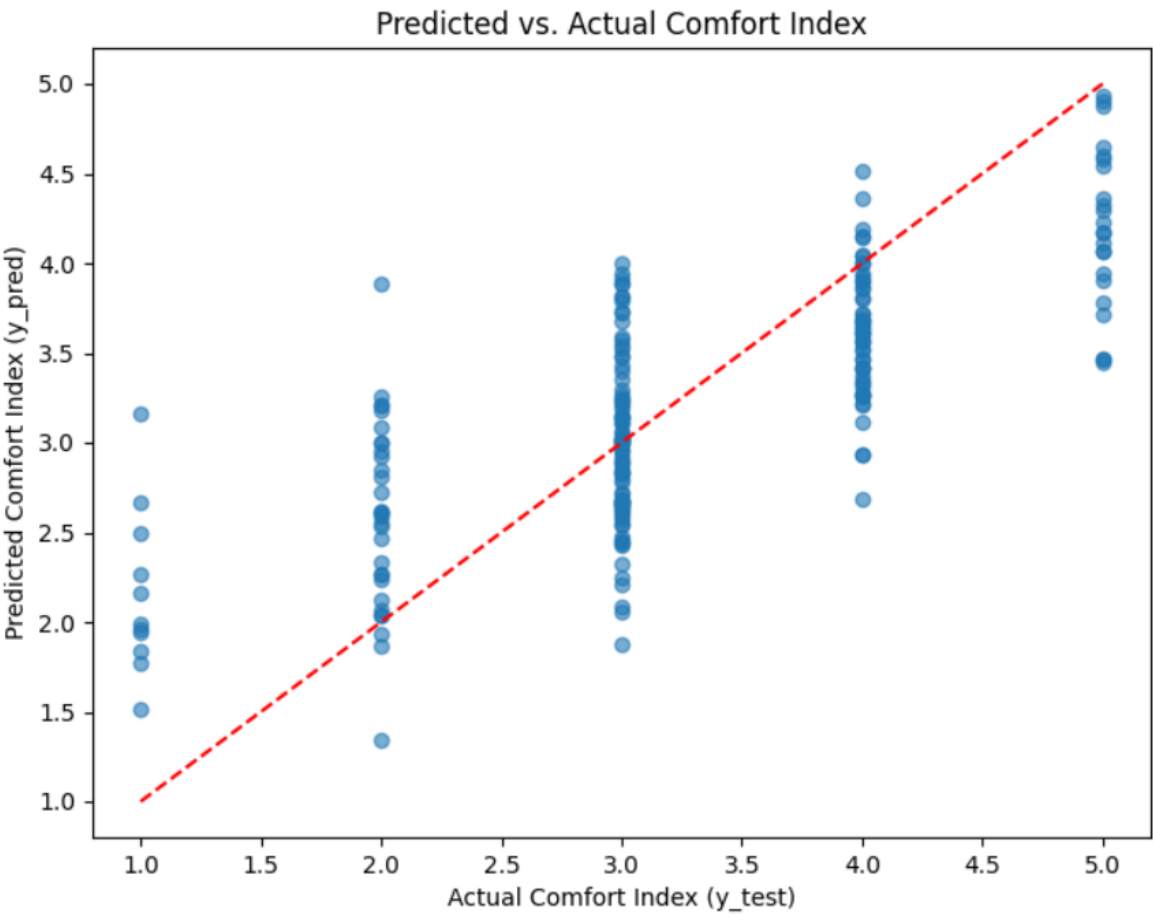


Figure 3 Actual vs. Predicted Comfort Pot

Table 16 Actual vs. Predicted Comfort Plot Description

Component	Technical Description
X-Axis (Actual Values)	Represents the true Comfort Index values generated during the synthetic data creation phase.
Y-Axis (Predicted Values)	Represents the Comfort Index values predicted by the trained model for the unseen test data.
Red Dashed Line (y=x)	Represents the " Perfect Prediction " line. If a point falls exactly on this line, the model's prediction matches the actual value perfectly.

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Category	Observation	Interpretation & Technical Insight
1. Clustering	Points are clustered in vertical columns.	This occurs because the synthetic ComfortIndex was constrained between 1 and 5 and rounded to the nearest integer (<code>round(0)</code>).
	Actual Comfort Index (y_test) consists only of integers.	The actual values are discrete (1.0, 2.0, 3.0, 4.0, 5.0) due to the rounding process.
	Predicted Comfort Index (y_pred) contains decimal values.	As a regression output, the model predicts continuous values (e.g., points between 3.0 and 4.0 represent predictions for a true value of 4.0).
2. Model Success	Majority of predictions are close to the red dashed line.	This indicates that the Random Forest Regressor effectively captures the general trend of the comfort index.
	Reasonable error margins.	For a true value of 3.0, most predictions fall within the 2.5–3.5 range, demonstrating acceptable variance.
3. Error Analysis	Strong performance at mid-range values (3.0, 4.0).	Predictions follow the identity line closely for moderate comfort levels.
	Boundary Effect at extreme values.	The model faces challenges at the edges of the scale (1.0 and 5.0).
	Overestimation at the lower bound (Actual 1.0).	For the lowest comfort level, the model tends to predict higher values (approx. 1.5–3.2).
	Underestimation at the upper bound (Actual 5.0).	For the highest comfort level, the model tends to predict lower values (approx. 3.5–4.5).

Figure 4 Analysis of the Plot

Table 17 Key Observations of the Plot

Key Observation	Technical Explanation
Relationship Learning	The model has successfully learned the underlying logic of your synthetic formula and captured the general correlation trend between environment and comfort.
Categorical Clustering	Because the actual Comfort Index consists of integers (1 to 5), the model attempts to produce results that closely mimic these discrete categories, even though it is a regression model.

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Key Observation	Technical Explanation
Boundary Deviation	Prediction deviations at the extremes (the tendency to predict 1.0 slightly higher and 5.0 slightly lower) are common in regression. This is often a result of the synthetic boundary constraints in your dataset.
Metric Consistency	Despite the clustering seen in the plot, your MSE and R² scores are expected to remain relatively strong, indicating a high degree of predictive power.

5.3. Feature Importance Ranking

The chart illustrates the relative weight (importance) the trained **Random Forest Regressor** assigns to each input variable when predicting the **Comfort Index**. In a Random Forest model, importance is calculated by measuring how much each feature contributes to reducing prediction error across all decision trees.

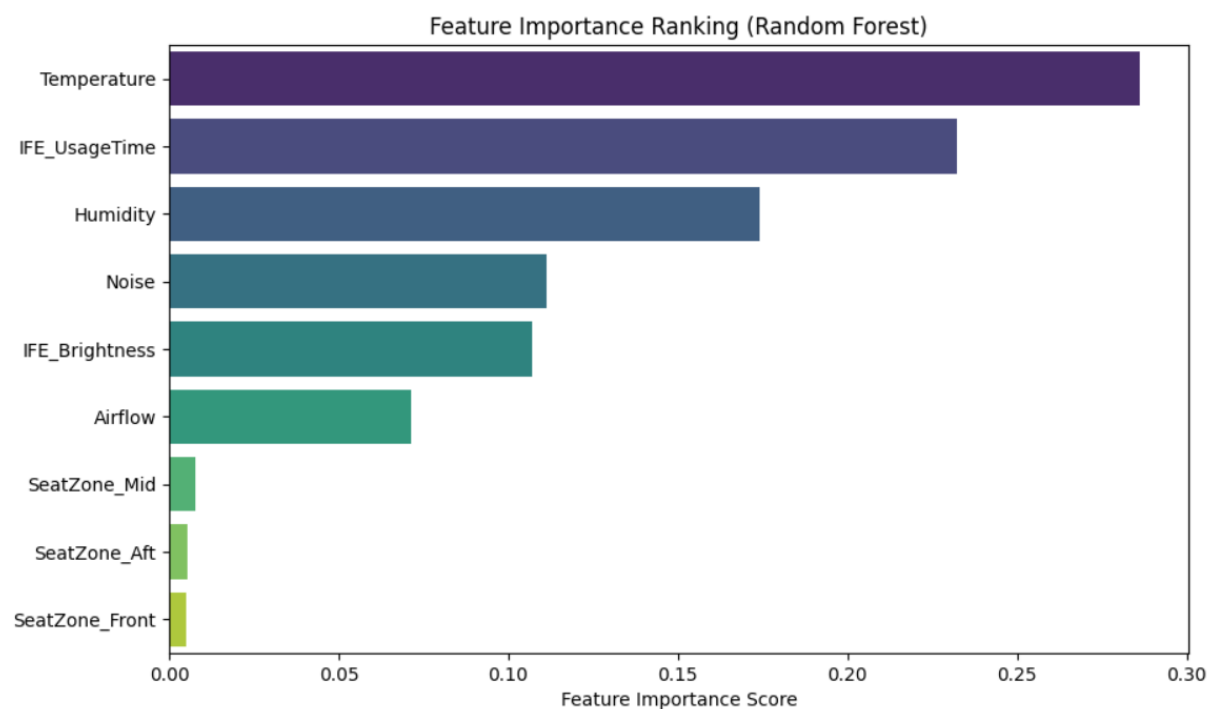


Figure 5 Feature Importance Ranking

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Table 18 Analysis for Feature Importance Ranking

Category	Feature	Importance Score (Approx.)	Interpretation & Insights
A. Primary Drivers	Temperature	0.29	Most critical feature. The model identified temperature as the primary determinant of comfort. This aligns with the high negative correlation (-0.45) found in the Heatmap.
	IFE_UsageTime	0.24	Second most important feature. The negative impact of prolonged IFE usage is perceived as the strongest factor after temperature, consistent with the moderate negative correlation (-0.40).
	Humidity	0.17	Key Non-linear Discovery. Although linear correlation was near 0.0, Random Forest successfully captured the non-linear relationship (due to the $\text{abs}(\text{Humidity} - 45)$ formula), identifying it as a critical factor.
B. Moderate Impact Features	Noise	0.11	Contributes meaningfully but is less dominant. The model determined that its impact on comfort is significant but secondary to the primary drivers.
	IFE_Brightness	0.1	Similar to noise, brightness affects comfort as expected, but plays a smaller role in the overall prediction compared to temperature or
	Airflow	0.07	The least critical among the continuous variables. Its effect on the final Comfort Index is relatively marginal.
C. Categorical Features	SeatZone (Front, Mid, Aft)	< 0.01	Negligible impact. After One-Hot Encoding, all seat zone variables showed near-zero importance. This confirms that seating location alone does not significantly influence comfort in this dataset.

5.4. Training Process and Final Performance Analysis

5.4.1. Hyperparameter Optimization

To find the most efficient version of the **Random Forest Regressor**, we used GridSearchCV. This process systematically tested different combinations of tree depth and quantity to minimize prediction error.

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Table 19 Hyperparameter Optimization

Parameter	Optimized Value	Technical Role
max_depth	10	Controls the complexity of the trees. A depth of 10 prevents the model from "memorizing" noise.
min_samples_leaf	2	Ensures each leaf has at least 2 samples, smoothing the predictions and reducing variance.
n_estimators	200	The forest consists of 200 individual trees, providing a robust ensemble average.

Table 20 Output of Hyperparameter Optimization

Output Line	Technical Description
Best Hyperparameters: <pre>{'max_depth': 10, 'min_samples_leaf': 2, 'n_estimators': 200}</pre>	GridSearchCV identified this specific combination as the most efficient for minimizing prediction error. The model was finalized using these settings: • n_estimators (200): The forest consists of 200 decision trees. More trees generally improve stability and performance. • max_depth (10): Limits tree depth to prevent the model from memorizing (overfitting) the training data.

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Output Line	Technical Description
	• min_samples_leaf (2): Ensures a minimum of 2 samples per leaf, smoothing the model's output.
Best MSE (Grid Search): 0.49337...	Represents the lowest average Mean Squared Error found during 3-fold Cross-Validation . This value indicates the model's internal performance during the training and validation cycles.

5.4.2. Final Model Evaluation

This section demonstrates the quality of predictions made by the best model found by GridSearchCV on the **reserved Test Set**—data the model has never seen before. This is the most reliable indicator of how well your model will perform in real-world scenarios.

Table 21 Final Model Evaluation

Performance Metric	Result	Interpretation
Mean Squared Error (MSE)	0.4450	On average, the model's predictions deviate by less than half a unit on the 1-5 scale.
R² Score (Determination)	0.5739	The model successfully explains 57.4% of the factors influencing passenger comfort.

5.4.3. Overall Model Performance Summary

The analysis confirms that the model is both robust and effective at capturing the underlying patterns of passenger comfort.

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Table 22 Overall Model Performance Summary

Metric	Result	Interpretation
R² Score	57.39%	Demonstrates moderately strong performance, confirming that predictions are significantly better than random guessing.
Test Set MSE	0.4450	Validates the model's accuracy on unseen data.
Training MSE	0.4933	The best average error recorded during the training phase.

- **Validation of Input Features:** The R² score validates that the selected input features have a substantial impact on passenger comfort.
- **No Overfitting:** The model shows no signs of overfitting.
- **Generalization:** Evidence of successful generalization is provided by the fact that the error on the test set (MSE: 0.4450) is lower than the best average error during the training phase (MSE: 0.4933).

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6. APPENDIX

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.ensemble import RandomForestRegressor
from sklearn.preprocessing import MinMaxScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.metrics import mean_squared_error, r2_score

# Sentetik Veri Oluşturma (NumPy)

np.random.seed(42)
N_samples = 1000

# Girdi Değişkenleri
Temperature = np.random.uniform(18, 26, N_samples) # °C
Humidity = np.random.uniform(20, 70, N_samples) # %
Noise = np.random.uniform(50, 80, N_samples) # dB
Airflow = np.random.uniform(0.5, 5, N_samples) # m³/h
SeatZone_map = {0: 'Front', 1: 'Mid', 2: 'Aft'}
SeatZone = np.random.randint(0, 3, N_samples)
IFE_Brightness = np.random.uniform(10, 100, N_samples) # %
IFE_UsageTime = np.random.uniform(0, 300, N_samples) # min

# Konfor İndeksi (1-5) için Basit Regresyon Formülü (sentetik)
# Düşük Sıcaklık, Orta Nem, Düşük Gürültü, Orta Hava Akımı, Düşük Parlaklık/Kullanım Süresi = Yüksek Konfor
ComfortIndex_raw = (
    5 - 0.2 * (Temperature - 22) # Sıcaklık 22'den uzaklaştıkça konfor düşer
    - 0.05 * abs(Humidity - 45) # Nem 45'ten uzaklaştıkça konfor düşer
    - 0.03 * (Noise - 50)
    + 0.1 * (Airflow - 2)
    - 0.01 * (IFE_Brightness - 50)
    - 0.005 * IFE_UsageTime
    + np.random.normal(0, 0.5, N_samples) # Gürültü/hata
)

```

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```

# Konfor İndeksi'ni 1-5 aralığına sıkıştırılma
ComfortIndex = np.clip(ComfortIndex_raw, 1, 5).round(0)

# DataFrame oluşturma
data = pd.DataFrame({
    'Temperature': Temperature,
    'Humidity': Humidity,
    'Noise': Noise,
    'Airflow': Airflow,
    'SeatZone': [SeatZone_map[z] for z in SeatZone],
    'IFE_Brightness': IFE_Brightness,
    'IFE_UsageTime': IFE_UsageTime,
    'ComfortIndex': ComfortIndex
})

print("Veri Setinin İlk 5 Satırı:")
print(data.head())

# Özellikler (X) ve Hedef (y) ayırma
X = data.drop('ComfortIndex', axis=1)
y = data['ComfortIndex']

# Eğitim ve Test setlerine ayırma
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Ön İşleme Hattı (Pipeline) Oluşturma
# Kategorik ve Sayısal Sütunları Tanımlama
categorical_features = ['SeatZone'] #
numerical_features = ['Temperature', 'Humidity', 'Noise', 'Airflow', 'IFE_Brightness', 'IFE_UsageTime'] #

# Dönüştürücüleri Tanımlama
preprocessor = ColumnTransformer(
    transformers=[
        ('num', MinMaxScaler(), numerical_features), # Normalizasyon (Min-Max)
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features) # One-hot encoding
    ],
    remainder='passthrough'
)

```

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```

# Ön İşlemeyi Uygulama
X_train_processed = preprocessor.fit_transform(X_train)
X_test_processed = preprocessor.transform(X_test)

# İşlenmiş veriyi DataFrame'e geri dönüştürme (gerekirse)
feature_names = numerical_features +
list(preprocessor.named_transformers_['cat'].get_feature_names_out(categorical_features))
X_train_df = pd.DataFrame(X_train_processed, columns=feature_names)
X_test_df = pd.DataFrame(X_test_processed, columns=feature_names)

# Sadece sayısal sütunları içeren bir DataFrame oluşturma
# 'SeatZone' sütununu ve One-Hot-Encoding yapılmamış kategorik sütunları çıkartma.

numerical_data = data.select_dtypes(include=np.number)

# Korelasyon Isı Haritası
plt.figure(figsize=(10, 8))
# Artık korelasyonu sadece sayısal veriler üzerinde hesaplıyoruz
sns.heatmap(numerical_data.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Korelasyon Isı Haritası (Sadece Sayısal Veriler)')
plt.show()

# Model Seçimi: Random Forest Regressor
rfr = RandomForestRegressor(random_state=42)

# Hiperparametre Optimizasyonu için Grid Search
param_grid = {
    'n_estimators': [50, 100, 200],
    'max_depth': [10, 20, None],
    'min_samples_leaf': [1, 2, 4]
}

grid_search = GridSearchCV(estimator=rfr, param_grid=param_grid, cv=3, scoring='neg_mean_squared_error',
n_jobs=-1, verbose=2)
grid_search.fit(X_train_df, y_train)

best_rfr = grid_search.best_estimator_

print("\nEn İyi Hiperparametreler:", grid_search.best_params_)
print("En İyi MSE (Grid Search):", -grid_search.best_score_)

```

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```

# Tahmin Yapma
y_pred = best_rfr.predict(X_test_df)

# Performans Metrikleri
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print(f"\nModel Performansı (Test Seti):")
print(f"Mean Squared Error (MSE): {mse:.4f}") # Minimize edilmesi gereken
print(f"R² Skoru: {r2:.4f}") # Maksimize edilmesi gereken

# 1. Tahmin Edilen vs. Gerçek Konfor Dağılım Grafiği
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.6)
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--') # Mükemmel Tahmin Çizgisi
plt.xlabel('Gerçek Konfor İndeksi (y_test)')
plt.ylabel('Tahmin Edilen Konfor İndeksi (y_pred)')
plt.title('Tahmin Edilen vs. Gerçek Konfor İndeksi')
plt.show()

# 2. Önemli Özellikler Grafiği (Feature Importance Plot)
if hasattr(best_rfr, 'feature_importances_'):
    feature_importances = pd.Series(best_rfr.feature_importances_, index=feature_names)
    feature_importances.sort_values(ascending=False, inplace=True)

    plt.figure(figsize=(10, 6))

    # !!! Düzeltilmiş Kod !!!
    # FutureWarning'ı çözmek için 'hue' parametresi eklendi ve 'legend=False' ayarlandı.
    sns.barplot(
        x=feature_importances.values,
        y=feature_importances.index,
        palette="viridis",
        hue=feature_importances.index, # Y eksenindeki değişkeni hue olarak atıyoruz
        legend=False # Göstergeyi kapatıyoruz
    )
    # !!! Düzeltilmiş Kod Sonu !!!

    plt.title('Özellik Önem Sıralaması (Random Forest)')
    plt.xlabel('Özellik Önemi Skoru')
    plt.ylabel('Özellik')
    plt.show()

```